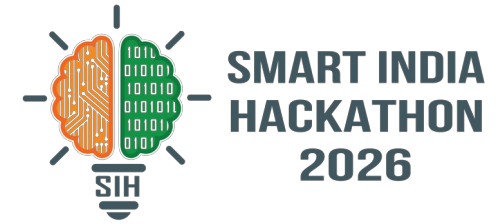


SMART INDIA HACKATHON 2026



Problem Statement ID - SIH26183

Problem Statement Title - Real-Time Identification of Fraud-Linked Cryptocurrency Exchanges from Victim-Reported Suspect Wallet Addresses through Automated Blockchain Analytics

Theme - Blockchain & Cybersecurity

PS Category - Software

Organisation - Ministry of Home Affairs

Team ID - 26183

Team Name (Registered on portal) - Avengers

Institute - Panipat Institute of Engineering and Technology



OUR SOLUTION

TraceFall

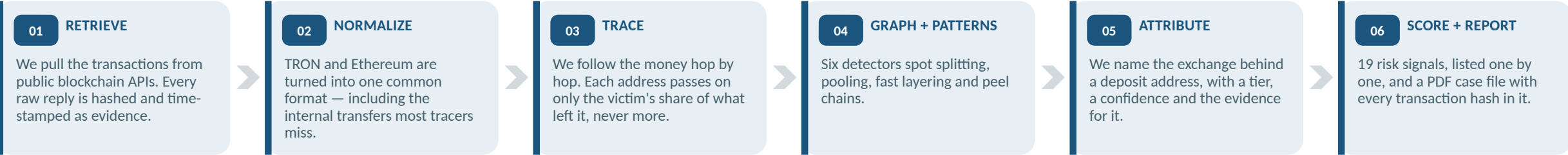
The address names no institution. The money does.



TraceFall — Which Exchange Received the Money?

A victim's wallet address goes in; the exchange that must receive the freeze request comes out, with the evidence attached.

PROPOSED SOLUTION (DESCRIBE YOUR IDEA/SOLUTION/PROTOTYPE) · DETAILED EXPLANATION OF THE PROPOSED SOLUTION



HOW IT ADDRESSES THE PROBLEM

AN INVESTIGATOR TODAY	WITH TRACEFALL
An address names nobody. A bank account number tells you the bank. A wallet address tells you nothing.	Behaviour names it. An address that collects from many strangers and sweeps it onward is a deposit address.
Two or three hops, by hand. The money splits faster than a person can follow it.	Every branch, automatically. The victim's share is carried correctly through each hop.
“Probably Binance.” A guess, with nothing behind it.	A tier and its evidence. CONFIRMED, PROBABLE or UNATTRIBUTED — never rounded up to certainty.
6–12 hours, usually “unknown”. Long after the money has moved on. (field estimate)	About 90 seconds on cached data. An evidenced answer to check, not a browser to click through.
■ CONFIRMED in a published dataset ■ PROBABLE worked out, with a confidence ■ UNATTRIBUTED we do not know, and we say so	

INNOVATION AND UNIQUENESS OF THE SOLUTION

- 01

THE THREE TIERS CAN NEVER BE MIXED UP

The database itself rejects any claim that is not CONFIRMED, PROBABLE or UNATTRIBUTED.
- 02

CONFIDENCE IS ONLY AS STRONG AS ITS WEAKEST LINK

$0.90 \text{ deposit} \times 0.80 \text{ exchange} = 0.72$. Never higher than that, and never above 0.95.
- 03

SEVEN NAMED REASONS A TRACE STOPS

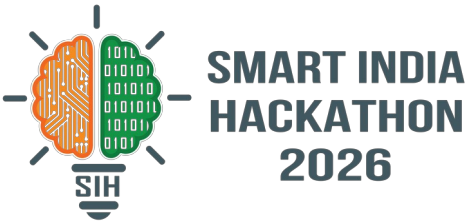
“We could not look further” is never reported as “the money stopped here”.
- 04

EVERY DETECTOR LISTS ITS OWN FALSE ALARMS

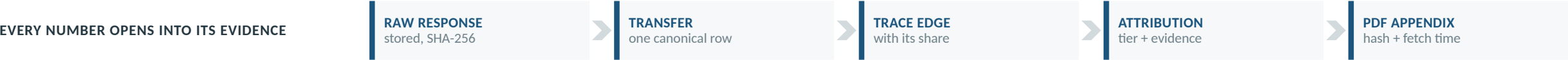
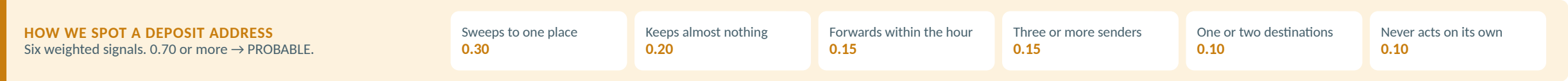
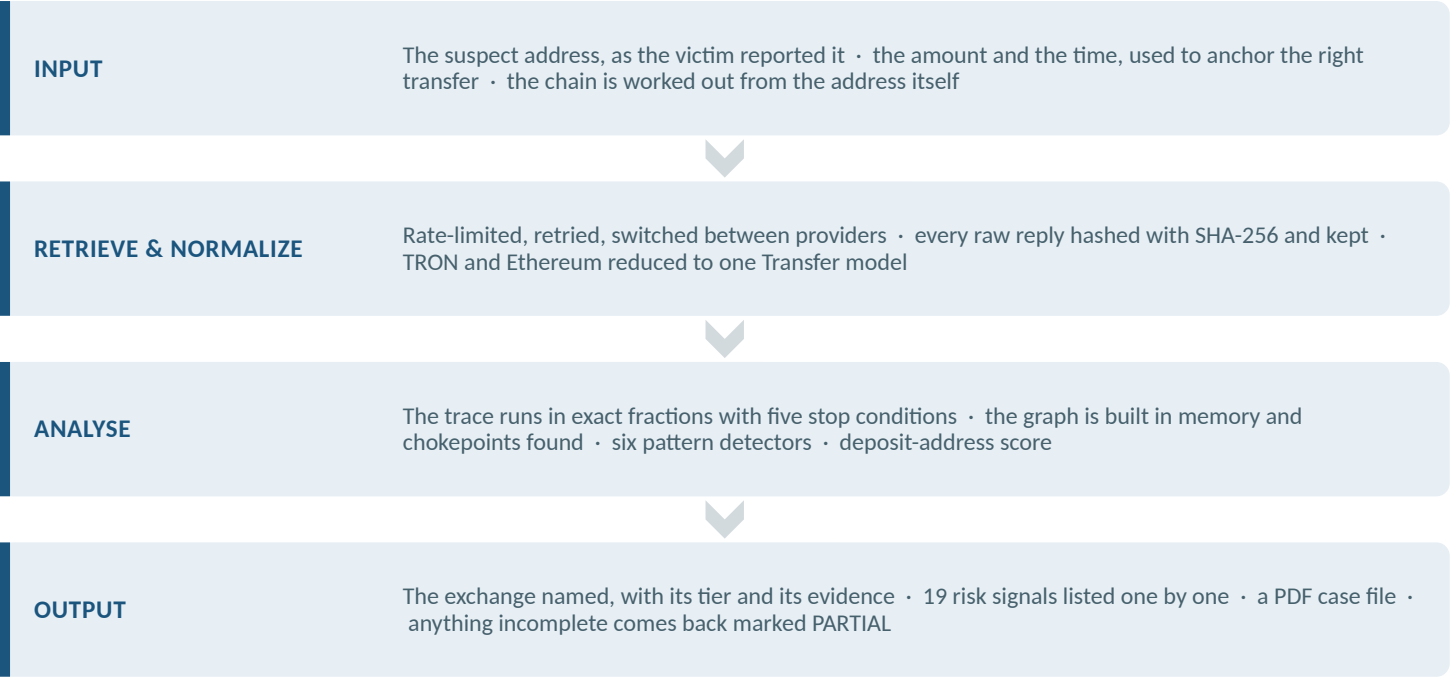
A detector that cannot name an innocent explanation is not allowed to run.



TECHNICAL APPROACH



METHODOLOGY AND PROCESS FOR IMPLEMENTATION (FLOW CHARTS/IMAGES/ WORKING PROTOTYPE)



TECHNOLOGIES USED

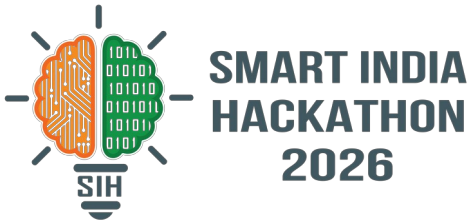
Frontend	React · TypeScript · Vite · Tailwind · Cytoscape.js
Backend	Python 3.12 · FastAPI · SQLAlchemy · Pydantic
Data	PostgreSQL 16 · Redis 7
Chain data	TronGrid · TronScan · Blockscout · Etherscan
Analytics & AI	NetworkX · LightGBM + SHAP (optional)
Reports & tests	ReportLab · pytest · Playwright

BUILT AND VERIFIED TODAY

- **Full pipeline runs:** retrieval → tracing → graph → patterns → attribution → risk → PDF
- **28 REST endpoints**, and a workspace with seven tabs and an interactive fund-flow graph
- **435 backend and 47 frontend tests** pass with no network access at all
- **94 source files**, clean under ruff and mypy strict
- **Append-only audit log** and evidence store, enforced by the database itself
- **Security headers** and per-IP rate limiting on every response
- **Not claimed:** the ML classifier is not built, and sign-in is single-factor



FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

IT ALREADY RUNS

An address goes in; a named exchange, an itemised risk score and a PDF come out. 482 tests pass with no internet.

NO PAID DATA NEEDED

Free public APIs and openly licensed datasets. No commercial forensics licence — which is why a district cyber cell could afford it.

ORDINARY DEPLOYMENT

One server: Python, PostgreSQL, Redis and a web app. No cluster, no GPU and no graph database to run.

THE SAME ANSWER EVERY TIME

Exact fractions, never floats. The same input gives the same answer — and every claim traces back to a stored, hashed API reply.

POTENTIAL CHALLENGES AND RISKS

TRON's free API is slow. We measured half a request per second per method, so a cold 200-address trace takes about 400 seconds — not 120.

Public TRON labels are few and dirty. Of 151 published exchange addresses we checked, 9 were not even valid TRON addresses.

A simple tracer cannot see internal transfers. On one sanctioned address, 173,600 ETH arrived where the ordinary transaction list showed nothing at all.

A behaviour test raises false alarms. Payment processors, OTC desks and company sweep accounts all look like deposit addresses.

A live demo depends on someone else's API. One rate-limit reply during a timed presentation ends the demo.

STRATEGIES FOR OVERCOMING THESE CHALLENGES

We work to an address budget, not to hope. About 60 new addresses inside 120 seconds; anything short of complete comes back marked PARTIAL. Cached and repeat traces stay fast.

We check every label on the chain ourselves. 94 of the 151 survived that check. Each one records its source, its licence and the date we captured it.

We fetch them and treat them like any other transfer. Without that step a tracer misses 95.2% of the money that came in, and draws its biggest arrow the wrong way round.

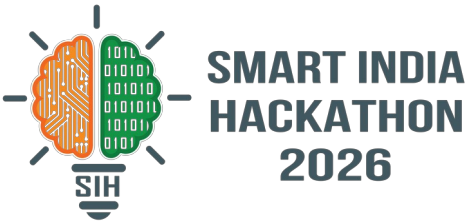
Behaviour alone never gives CONFIRMED, and a detector that cannot name an innocent explanation for the shape it flags is not allowed to run.

Offline by default. Real chain data, captured and frozen, replayed from disk behind a visible “cached snapshot” banner. The whole suite runs with no network.

Every figure above is our own measurement, taken on 2026-09-05 and written up in the repository — not an estimate.



IMPACT AND BENEFITS



POTENTIAL IMPACT ON THE TARGET AUDIENCE

22,68,346

cybercrime incidents on NCRP in 2024, up 42% on 2023

₹22,845.73 cr

reported lost to cybercrime in 2024

₹7,130 cr

saved through CFCFRMS across 23.02 lakh complaints

Ministry of Home Affairs, Lok Sabha Unstarred Question 432, answered 2 December 2025

PRIMARY USER DISTRICT / STATE CYBER CELL INVESTIGATOR

Not a blockchain expert, and does not need to become one. Types in the address the victim reported and gets back a named institution to serve, the hops in between, and the hashes to attach to the request.

SECONDARY I4C / CENTRAL ANALYST

Sees the same deposit address turn up in complaints from different states — which turns twenty unconnected case files into one operation.

SECONDARY SUPERVISORY OFFICER

Sorts a queue by where the money still seems to be, and can read why any finding was made without having to ask an engineer.

BENEFITS OF THE SOLUTION (SOCIAL, ECONOMIC, ENVIRONMENTAL, ETC.)

HOW IT IS DONE TODAY

A block explorer is opened and clicked through two or three hops by hand. Six to twelve hours per address, and it usually ends in “unknown”.

Field estimate. No measured study exists, and we do not claim one.



WITH TRACEFALL

The automated stages finish in about ninety seconds on cached or warm data. The investigator checks an evidenced answer instead of producing one.

Measured on captured chain data. A cold live trace is limited by provider rate limits, and says so on screen.

REACHES THE EXCHANGE IN TIME

A freeze request needs someone to send it to. Naming that exchange early is the difference between getting the money back and not.

FINDS VICTIMS WHO NEVER COMPLAINED

Tracing backwards from a scam wallet lists the other addresses that paid into it — people who never filed a complaint.

NO PER-SEAT LICENCE COST

Commercial forensics platforms cost more than most state cyber cells can pay. This runs on free public data, on one server.

AN ANSWER THAT CAN BE CHECKED

Every number opens into its signals, its transactions, and the stored raw reply with its hash and the time it was fetched.

WHERE THIS SYSTEM STOPS

Said by us, before anyone asks.

It does not name a person. Public blockchain data cannot do that, and we will not pretend otherwise.

It does not decide that fraud happened. A risk score is an argument built from named signals — not a probability of guilt.

It is not plugged into NCRP or CFCFRMS. We offer an integration-ready API and claim nothing more than that.

DETAILS / LINKS OF THE REFERENCE AND RESEARCH WORK

GOVERNMENT MANDATE, SCALE AND LAW

MHA / I4C — SOP for NCRP-CFCFRMS, 2 January 2026

Where funds reach an exchange not onboarded to CFCFRMS, the officer “may conduct a VDA forensic analysis ... to ascertain that which VASP or VDA Exchange has control of the wallet.” Issuance confirmed in Rajya Sabha USQ 553, 4 Feb 2026.

mha.gov.in/MHA1/Par2017/pdfs/par2026-pdfs/RS04022026/553.pdf

MHA — Lok Sabha Unstarred Q. 432, 2 December 2025

22,68,346 NCRP incidents in 2024 (+42.08%); ₹22,845.73 crore reported lost; ₹7,130 crore saved via CFCFRMS.

mha.gov.in/MHA1/Par2017/pdfs/par2025-pdfs/LS02122025/432.pdf

Standing Committee on Home Affairs — 254th Report, 20 Aug 2025

Recommends training in “digital forensics, blockchain analysis and virtual asset tracing”.

sansad.in/Committee_File/ReportFile/254_2025_8_12.pdf

Ministry of Finance — Gazette S.O. 1072(E), 7 March 2023

Brings virtual digital asset services under the PMLA, 2002.

egazette.gov.in/WriteReadData/2023/244184.pdf

PIB / FIU-IND, 1 October 2025

50 VDA service providers registered; non-compliance notices issued to 25 offshore exchanges.

pib.gov.in/PressReleasePage.aspx?PRID=2173758

WHY TRON AND USDT

UNODC — Casinos, Money Laundering, Underground Banking and Transnational Organized Crime in East and Southeast Asia, Jan 2024

“USDT on the TRON blockchain has become a preferred choice for regional cyberfraud operations and money launderers alike due to its stability and the ease, anonymity, and low fees of its transactions.” (p. 50) The same page reports an audit finding 17.07 billion USDT tied to underground exchanges and criminal activity between Sept 2022 and Sept 2023.

unodc.org/roseap/Casino_Underground_Banking_Report_2024.pdf

TRM Labs — 2025 Crypto Crime Report, 10 February 2025

58% of illicit crypto volume in 2024 was on TRON, ahead of Ethereum (24%) and Bitcoin (12%). Illicit volume overall, not investment fraud alone.

trmlabs.com/reports-and-whitepapers/2025-crypto-crime-report

Enforcement Directorate — press release, 20 November 2025

Proceeds of a ₹285-crore cyber-fraud network turned into USDT over exchange P2P using non-KYC accounts; ₹8.46 crore attached across 92 accounts. Names USDT; the chain is not stated.

enforcementdirectorate.gov.in/PressRelease-PAO-Cyber-Fraud-Case

Enforcement Directorate — press release, 26 September 2025

An investment scam promising 5–6% monthly returns collected funds through payment aggregators and USDT; ₹391 crore attached across 185 accounts.

enforcementdirectorate.gov.in/PressRelease-Navab-Hassan-QFX

DATA SOURCES AND OUR OWN MEASUREMENTS

TronGrid • Blockscout • Etherscan

Transaction data for TRON and Ethereum, used under their API terms. We never scrape explorer label databases.

developers.tron.network/reference/rate-limits • docs.blockscout.com

OFAC SDN sanctioned digital-currency addresses

405 addresses ingested (281 TRON, 124 Ethereum), extracted with the MIT-licensed 0xB10C tool.

github.com/0xB10C/ofac-sanctioned-digital-currency-addresses

Dune spellbook — TRON exchange addresses

A lead list only. Its BSL 1.1 licence does not cover our use, so we re-establish every address ourselves and never load theirs in.

github.com/duneanalytics/spellbook

Our measurements, 2026-09-05 (written up in the repository)

TronGrid sustains half a request per second per method; Blockscout took 195 requests without throttling. 9 of 151 published TRON exchange labels are not valid addresses; 94 survived our checks.

docs/research/OQ-01-provider-rate-limits.md • [OQ-08-tron-label-coverage.md](https://docs/OQ-08-tron-label-coverage.md)

Our design record — 21 architecture decisions and a written limits list

Why TRON and Ethereum first; why value is shared out in proportion; why there is no graph database; why the database enforces the three tiers; and what we will not claim.

docs/DECISIONS.md • docs/LIMITATIONS.md

HOW WE USE THESE SOURCES

We never scrape explorer label databases ? their terms forbid it. A dataset whose licence does not allow our use is treated as a lead to check ourselves, never loaded in. Every label we ship records its source, its licence and the date we captured it.